CORTEX — Clinical Evaluation Report (CER) Module

Module Specification v1.2 — February 2026

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1. Overview

The CER module is the assembly and narrative engine of the CORTEX Clinical Affairs suite. It consumes the outputs of every upstream module — SLS (literature evidence), SOA (structured analysis, benchmarks, claims, quality assessments), and Validation (device performance, clinical benefit, GSPR mapping) — and assembles them into a regulatory-grade Clinical Evaluation Report compliant with MDR Annex XIV, MEDDEV 2.7/1 Rev 4, and MDCG 2020-13.

The CER module produces the **20.CER** document for MDR submissions. For FDA submissions, clinical validation evidence is documented in the standalone **18.CVS** (Clinical Validation Study), produced by the Validation module. The 20.CER addresses MDR requirements only and references MDR validation studies as appendices. The 18.CVS is a parallel deliverable, not an appendix of the CER.

The CER is not a copy-paste aggregation. It is an authored document with a continuous narrative thread, cross-referenced claims, and a structured benefit-risk determination. The module provides AI-assisted drafting, human review workflows, version management, and export to submission-ready DOCX.

Core principles:

- Every statement in the CER must be traceable to its source (SLS article, SOA section, Validation result, Risk Management file)
- The CER is a living document — it is versioned, and each version locks its upstream dependencies
- AI drafts, humans review and finalize — the AI never publishes autonomously
- The CEP (Clinical Evaluation Plan) is not a separate module: it is the CER in `planned` status, exported with prospective language
- The PCCP (Performance and Clinical Claims Plan) is a companion planning document generated from the same data model, documenting intended claims before validation

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2. Upstream Dependencies

```
SLS (locked)        SOA Clinical (locked)  SOA Device (locked)     Validation (completed/locked)
│                   │                      │                       │
│ search strategy   │ clinical context     │ benchmarks              │ device performance
│ PRISMA flowchart  │ sections & claims    │ comparison tables       │ clinical benefit (MRMC)
│ article list      │ similar device       │ equivalence argument    │ GSPR mapping matrix
│ exclusion summary │ registry             │ device claims           │ benefit quantification
```

```
│              │ quality assessments │                │ fairness assessment
▼              ▼                ▼                ▼
┌──────────────────────────────────────────────────────────────────────────────────────┐
│                        CER MODULE (20.CER)                                           │
│                                                                                      │
│  + External Document References (Risk Management, Usability, IFU, etc.)              │
│  + Named Device Literature Search (SLS session, type: named_device)                  │
│  + PMS / PMCF summary (from PMS module)                                              │
│                                                                                      │
│  → Outputs: 20.CER (DOCX), CEP (DOCX), PCCP (DOCX),                                  │
│          GSPR Table, Benefit-Risk Summary                                            │
└──────────────────────────────────────────────────────────────────────────────────────┘

Parallel FDA path (NOT part of this module):
  SOA Device (shared) → Validation [FDA context] → 18.CVS (standalone)
```

External documents (not produced by CORTEX but referenced in the CER):

- Risk Management Report & Risk Analysis Table
- Security Risk Management Plan
- Usability Summative Evaluation
- Software User Manual / IFU
- Project Master Plan
- Methods Document

These are referenced and summarized — not imported in full. The CER contains a narrative summary of each, written or AI-drafted from a user-provided abstract.

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3. Data Model

3.1 Hierarchy

```
Project
├─ CER Analysis
│  ├─ Configuration (version, linked modules, regulatory context)
│  ├─ External Document References (list of referenced docs with summaries)
│  ├─ Section Template (configurable CER structure)
│  ├─ CER Sections (ordered, with narrative + source references)
│  │  └─ Section Claims (linked to SOA/Validation claims)
│  ├─ GSPR Compliance Matrix (linked to Validation GSPR mapping)
│  ├─ Benefit-Risk Determination
│  │  ├─ Benefit Registry (from Validation)
```

```
│  │  └─ Risk Summary (from external Risk Management doc)
│  ├─ PCCP Record (planned claims snapshot)
│  ├─ Bibliography (auto-compiled from all referenced articles)
│  └─ Evaluator Records (persons involved in writing/review/approval)
```

3.2 CER Analysis Record

Field	Description
id	UUID
project_id	Parent project
name	e.g., "CER CINA-CSpine v1.0.1"
version	Document version (e.g., "01", "02")
version_reason	Why this version was created (e.g., "Initial CE marking", "Annual PMS update", "Patch v1.0.2")
version_type	initial | annual_update | patch_update
status	planned → drafting → review → completed → locked
linked_soa_clinical_id	Locked SOA Clinical analysis
linked_soa_device_id	Locked SOA Device analysis (nullable)
linked_validation_ids	One or more completed/locked Validation studies (MDR context only)
linked_sls_named_device_id	SLS session for named device literature (nullable)
linked_pms_cycle_id	PMS cycle providing post-market data (nullable — initial CER has none)
parent_cer_id	For updates: reference to the previous CER version (nullable)
regulatory_context	CE-MDR (this module only produces MDR CERs)
device_classification	e.g., "IIb, rule 11, MDCG 2019-11 category III.i"
imdrf_samd_category	SaMD risk categorization per IMDRF framework (ref: AV-DP-CINA-10-022-EU-classif EU(MDR))
update_frequency	e.g., "1 year" — pulled from project config
benefit_risk_framework	mdr_bl_rl (this module uses MDR scored BL/RL framework)
notes	Free-form

Note on IMDRF SaMD classification: The `imdrf_samd_category` field references the SaMD risk categorization per IMDRF/SaMD WG/N12FINAL:2014, as documented in Avicenna.AI's EU classification document (AV-DP-CINA-10-022-EU-classif EU(MDR)). This classification informs the required level of clinical evidence and is referenced in the CER scope section. This is distinct from the IMDRF Adverse Event Terminology (IMDRF/AE WG/N43) used for complaint/incident classification in the PMS module.

3.3 CEP / CER Duality

The Clinical Evaluation Plan (CEP) and the Clinical Evaluation Report (CER) share identical structure. The CEP is the CER when:

- The CER status is `planned`
- Sections describe what will be done (prospective language)
- Data fields contain targets and TBDs rather than results

Implementation:

- When `status = planned`, the export generates a **CEP** document (prospective language, TBD placeholders)
- When `status ≥ drafting`, the export generates a **CER** document (past tense, actual data)
- The system maintains both views from the same underlying data
- AI drafting adapts its language based on the CER status:
  - `planned`: "Avicenna.AI will conduct a retrospective study..."
  - `drafting`+: "Avicenna.AI conducted a retrospective study..."

CEP-specific fields (shown only when `status = planned`):

- For each section: `planned_methodology` (text describing the planned approach)
- For claims: `target_values` (e.g., "Sensitivity ≥ 60%") instead of `achieved_values`
- For benefit-risk: `expected_bl` and `expected_rl` instead of computed values

When the CER transitions from `planned` to `drafting`, the CEP fields are archived (for audit) and the CER fields are populated from upstream modules.

3.4 PCCP (Performance and Clinical Claims Plan)

The PCCP is a companion planning document that formalizes intended performance and clinical benefit claims before validation. It creates a traceable link between SOA-derived targets, planned validation endpoints, and final CER claims.

PCCP Record:

Field	Description
id	UUID
cer_analysis_id	Parent CER analysis
version	Document version
status	draft → reviewed → approved → superseded
claims_snapshot	JSON — snapshot of all claims at time of approval
approved_by	User who approved
approved_at	Timestamp

Key behaviors:

- The PCCP is created during the CER planned phase (alongside the CEP)
- When the CER transitions to drafting, the PCCP is archived as a baseline
- Post-validation, a comparison view shows: PCCP planned claims vs. CER achieved claims
- Deviations between PCCP targets and actual results are flagged and require documented justification in the CER Discussion section
- Export as DOCX with Avicenna.AI formatting

PCCP Section Template:

```
1. INTRODUCTION
  ├── 1.1 Document Overview
  ├── 1.2 Device Description (auto-populated from project)
  └── 1.3 Regulatory Context

2. INTENDED PERFORMANCE CLAIMS
  ├── 2.1 Technical Performance Claims
  │   Source: SOA Device → benchmarks → acceptance criteria
  │   Content: each claim with target value, SOA justification,
  │   planned validation endpoint mapping
  │
  └── 2.2 Performance Acceptance Criteria
      Source: Validation protocol → primary/secondary endpoints
      Content: acceptance criteria with statistical methods and
      sample size justification

3. INTENDED CLINICAL BENEFIT CLAIMS
  ├── 3.1 Clinical Benefit Claims
  │   Source: SOA Clinical + SOA Device → clinical benefit sections
  │   Content: each claimed benefit with SOA evidence basis
  │
  └── 3.2 Clinical Benefit Validation Strategy
      Source: Validation protocol → MRMC design
      Content: how each benefit will be demonstrated

4. SAFETY AND RISK CLAIMS
  ├── 4.1 Known Risk Profile
  │   Source: External Risk Management summary
  │
  └── 4.2 Risk Monitoring Strategy
      Source: PMS/PMCF Plan → gap registry

5. CLAIM TRACEABILITY MATRIX
  Content: Claims × SOA Evidence × Validation Endpoint ×
  CER Section mapping (auto-generated)

6. UPDATE SCHEDULE
  Content: When the PCCP will be reviewed
  (aligned with CER update frequency)
```

PCCP Traceability Matrix (auto-generated):

Claim ID	Claim Text	SOA Source	Validation Endpoint	CER Section	Status
PC-01	Sensitivity ≥ 60%	SOA Device §2.2 Benchmark	Standalone: Primary EP	CER §3.4.1	Planned → Achieved: 90.3%
PC-02	Reading time reduction	SOA Device §3.2	MRMC: Primary EP	CER §3.4.2	Planned → Achieved: 35.7%
PC-03	No significant bias across scanner makers	SOA Device §5.1	Fairness Assessment	CER §5.2.3.1.2	Planned → Confirmed

4. Section Template

4.1 Concept

The CER follows a structured template matching MEDDEV 2.7/1 Rev 4 and the actual document format used by Avicenna.AI. Sections are ordered, nested, and each maps to specific upstream data sources.

4.2 Section Record

Field	Description
id	UUID
cer_analysis_id	Parent CER
key	Machine-readable (e.g., `scope.device_description`)
title	Display title (e.g., "3.2. Description of the Device")
content	Rich text narrative (Markdown) — AI-generated + human-edited
content_cep	Rich text for CEP version (prospective language) — maintained separately
source_type	`soa_clinical` | `soa_device` | `validation` | `risk_mgmt` | `usability` | `pms` | `manual` | `computed`
source_module_ids	Which upstream module sections feed this CER section
source_article_ids	Articles directly referenced in this section
status	`empty` → `planned` → `drafted` → `reviewed` → `final`
drafted_by	`ai` | `manual`

Field	Description
reviewed_by	User who reviewed
sort_order	Position in the CER
parent_section_key	For sub-sections

4.3 Default Section Template

```
1. GENERAL INFORMATION
   │ Source: project definition (device description, classification,
   │ regulatory status, IMDRF SaMD category)
   │ Mode: auto-populated from project fields, editable
   │

2. GENERAL SAFETY AND PERFORMANCE REQUIREMENTS (GSPR)
   │ Source: Validation → GSPR mapping matrix
   │ Mode: auto-populated table + narrative wrapper
   │
   ├── GSPR Table (auto-populated)
   └── Narrative: "The identified GSPRs that require support
     from clinical data are..."

3. SCOPE OF THE CLINICAL EVALUATION
   ├── 3.1 Name and Address of the Manufacturer
   │   Source: project config
   │   Mode: auto-populated
   │
   ├── 3.2 Description of the Device
   │   ├── 3.2.1 General Description
   │   │   Source: project definition (device table with EMDN, UDI,
   │   │   classification, novelty, IMDRF SaMD category)
   │   │   Mode: auto-populated table + narrative
   │   │
   │   ├── 3.2.2 Intended Purpose
   │   │   Source: project definition (intended use, population,
   │   │   contraindications, etc.)
   │   │   Mode: auto-populated table
   │   │
   │   └── 3.2.3 Detailed Description
   │       Source: project definition (technical description)
   │       Mode: auto-populated table
   │
   ├── 3.3 Description of the Situation
   │   Source: project definition (product history, equivalence,
```

```
│   alternatives)
│   Mode: auto-populated tables + narrative
│
└── 3.4 Claims
    ├── 3.4.1 Intended Technical Performances
    │   Source: SOA Device → benchmark claims + acceptance criteria
    │   + PCCP planned claims (when available)
    │   Mode: AI-drafted from SOA claims
    │
    ├── 3.4.2 Intended Clinical Benefits
    │   Source: SOA Device → clinical benefit claims + Validation →
    │   benefit registry + PCCP planned claims
    │   Mode: AI-drafted from SOA + Validation
    │
    └── 3.4.3 Clinical Safety
        Source: external Risk Management summary + SOA → safety claims
        Mode: AI-drafted from risk summary

4. CLINICAL EVALUATION STRATEGY
  │ This section describes the overall evaluation methodology.
  │ In CEP mode: prospective ("will be conducted")
  │ In CER mode: past tense ("was conducted")
  │
  ├── 4.1 General Strategy
  │   ├── 4.1.1 Elements from the Pre-market Phase
  │   │   ├── 4.1.1.1 State of the Art
  │   │   │   Source: SOA Clinical + SOA Device → summary
  │   │   │   Mode: AI-drafted summary of SOA scope and approach
  │   │   │
  │   │   ├── 4.1.1.2 Preclinical Technical Performance Studies
  │   │   │   Source: Validation → standalone study summary
  │   │   │   Mode: AI-drafted summary of study design
  │   │   │
  │   │   ├── 4.1.1.3 Clinical Benefit Studies
  │   │   │   Source: Validation → MRMC study summary
  │   │   │   Mode: AI-drafted summary of study design
  │   │   │
  │   │   ├── 4.1.1.4 Summative Usability Testing
  │   │   │   Source: external Usability document → user-provided summary
  │   │   │   Mode: manual or AI-drafted from provided text
  │   │   │
  │   │   └── 4.1.1.5 Risks
  │   │       Source: external Risk Management Plan reference
  │   │       Mode: manual reference
  │   │
  │   ├── 4.1.2 Elements from the Post-market Phase
  │   │   Source: PMS module → summary or placeholder
```

```
│   │   Mode: AI-drafted or placeholder (initial CER)
│   │
│   └── 4.1.3 Compilation of Elements
│       Source: manual
│       Mode: standard boilerplate, editable
│
└── 4.2 Equivalence Strategy
    Source: SOA Device → equivalence argumentation (or "no equivalence")
    Mode: AI-drafted from SOA equivalence section or manual statement

5. SOURCES OF CLINICAL DATA
   │ This is the core body of the CER — the largest section.
   │
   ├── 5.1 The State of the Art (SOA)
   │   ├── 5.1.1 Clinical Background and Current Knowledge
   │   │   Source: SOA Clinical → thematic sections (§1-§5)
   │   │   Mode: AI-drafted synthesis from SOA clinical sections
   │   │   Note: NOT a verbatim copy — the CER summarizes
   │   │   and contextualizes
   │   │
   │   └── 5.1.2 Clinical Data from Literature Related to Similar Devices
   │       Source: SOA Device → thematic sections + benchmarks
   │       + comparison tables
   │       Mode: AI-drafted synthesis from SOA device sections
   │       Includes: performance ranges, clinical benefit data
   │       from SOA, limitations
   │
   ├── 5.2 Clinical Data Generated and Held by the Manufacturer
   │   ├── 5.2.1 Risk Management Analysis
   │   │   Source: external Risk Management Report →
   │   │   user-provided summary
   │   │   Mode: manual or AI-drafted from provided text
   │   │   Content: risk count summary, RL levels,
   │   │   residual risk assessment
   │   │
   │   ├── 5.2.2 Verification Test Results
   │   │   ├── 5.2.2.1 Formative Usability Testing
   │   │   │   Source: manual
   │   │   │
   │   │   └── 5.2.2.2 Summative Usability Testing
   │   │       Source: external Usability Report →
   │   │       user-provided summary
   │   │       Mode: manual or AI-drafted from provided text
   │   │
   │   └── 5.2.3 Clinical Development Report
   │       ├── 5.2.3.1 Pre-clinical Studies
   │       │   ├── 5.2.3.1.1 Pilot Technical Performance Testing
```

```
│       │   │   (optional)
│       │   │   Source: manual (historical pilot data)
│       │   │
│       │   ├── 5.2.3.1.2 Pivotal Standalone Validation Study
│       │   │   Source: Validation module → standalone study
│       │   │   Mode: AI-drafted from Validation data
│       │   │   Sub-sections auto-generated:
│       │   │   a) PRIMARY ENDPOINT
│       │   │   b) SECONDARY ENDPOINT
│       │   │   c) ADDITIONAL SUPPORTIVE STATISTICS
│       │   │   d) ADDITIONAL ASSESSMENTS (if applicable)
│       │   │   e) CONCLUSION
│       │   │
│       │   └── 5.2.3.1.3 Clinical Benefit Study (MRMC)
│       │       Source: Validation module → MRMC study
│       │       Mode: AI-drafted from Validation data
│       │       Content: study design, reader table,
│       │       endpoints, results, conclusion
│       │
│       └── 5.2.3.2 Post-Market Data
│           ├── 5.2.3.2.1 PMCF
│           │   Source: PMS module → PMCF summary
│           │   (or placeholder for initial CER)
│           │   Mode: placeholder for initial CER;
│           │   AI-drafted from PMS data for updates
│           │
│           └── 5.2.3.2.2 PMS / Vigilance
│               Source: PMS module → PSUR summary
│               (or placeholder for initial CER)
│               Mode: placeholder for initial CER;
│               AI-drafted from PMS data for updates
│
└── 5.3 Summary of Published Literature Concerning the Named Device
    Source: SLS session (type: named_device) → results summary
    Mode: AI-drafted from SLS data (search strategy, PRISMA,
    included articles or "no data found")

6. CRITERIA FOR ACCEPTABILITY OF THE BENEFIT/RISK RATIO
   │ Source: Validation → benefit quantification + external
   │ Risk Management
   │ Mode: structured computation + AI-drafted narrative
   │ Framework: MDR BL/RL scored matrix
   │
   │ Note: This section uses the MDR benefit-risk framework
   │ (Benefit Level × Risk Level scoring). FDA submissions
   │ use a qualitative benefit-risk assessment per FDA's
   │ Benefit-Risk Guidance — this is handled in the 18.CVS,
```

```
│ not in the CER. CORTEX does not map between the two
│ frameworks.
│
├── Benefit-Risk Summary Table (auto-populated)
│   Left column: benefits with BL from Validation
│   Right column: risks with RL from Risk Management
│   (user-entered)
│
├── Magnitude Scale Table (configurable, pre-filled with
│   default 5-point scale)
│
├── 6.1 Benefit #B1 Analysis (one sub-section per benefit)
│   Source: Validation → benefit registry + SOA context
│   Mode: AI-drafted narrative incorporating:
│   - Clinical context from SOA (why this benefit matters)
│   - Evidence from Validation (statistical results)
│   - Probability and magnitude scoring with justification
│   - BL calculation
│   - PCCP comparison (if available): planned vs. achieved
│
├── 6.N Benefit #BN Analysis (repeatable)
│
└── 6.X Benefit/Risk Conclusion
    Source: computed BL vs RL + AI-drafted narrative
    Mode: auto-computed ratio + AI-drafted conclusion
    Content: "BL = X outweighs RL = Y", off-label RL
    assessment, final determination

7. GENERAL DISCUSSION AND CONCLUSIONS
  Source: all upstream modules
  Mode: AI-drafted synthesis
  Content: comprehensive summary of all evidence,
  GSPR compliance statement, PCCP deviation summary
  (if applicable), forward-looking statement on PMS/PMCF

8. CER UPDATE FREQUENCY
  Source: project config
  Mode: auto-populated with standard text

9. QUALIFICATION OF THE RESPONSIBLE EVALUATORS
  Source: evaluator records (see 4.7)
  Mode: auto-populated table + CV references
```

4.4 Section Customization

- Users can add custom sections at any level (parent or sub-section)

- Users can hide/remove default sections (hidden sections retain their data)
- Users can rename section titles
- Users can reorder sections via drag & drop
- Custom section templates can be saved for reuse across projects
- Sections 5.2.3.1.2 and 5.2.3.1.3 (Validation results) auto-adapt their sub-structure based on the linked Validation study type (standalone, MRMC, combined)

4.5 AI Section Drafting

For each section, the user can click "Draft this section":

1. **Input to LLM:** section title + description + source data depending on `source_type`:
   - `soa_clinical` / `soa_device`: relevant SOA section content + claims + article references
   - `validation`: protocol data + imported results + endpoint mapping + pass/fail status
   - `risk_mgmt` / `usability`: user-provided summary text for the external document
   - `pms`: PMS cycle data (surveys summary, vigilance summary, literature update)
   - `computed`: structured data (GSPR matrix, benefit-risk calculation)
2. **CER-specific drafting instructions:**
   - The CER summarizes and contextualizes — it does not copy verbatim from SOA or Validation
   - The AI synthesizes across sources: e.g., section 5.1.1 draws from multiple SOA Clinical sections
   - Inline article references formatted as `[Author, Year]` or `[reference number]`
   - Cross-references to appendices: e.g., "as detailed in the Standalone Validation Report [R2]"
   - Language adapts to CER status: prospective for `planned`, past tense for `drafting`+
   - PCCP context: when a PCCP is linked, the AI references planned claims and highlights any deviations in the Discussion section
3. **Output:** Markdown narrative draft populating the section's `content` field
4. **Regenerate button** available with optional guidance ("focus on the comparison with BriefCase", "more concise")
5. Section status moves from `empty` to `drafted`

4.6 External Document References

The CER references several documents not produced by CORTEX. These are managed as a structured list:

Field	Description
id	UUID
cer_analysis_id	Parent CER
ref_code	e.g., "[R6]"
document_identifier	e.g., "AV-DP-CINA-CSP-10-005-annex2-RISK-REPORT-V02"
document_title	e.g., "Risk Management Report"
document_type	risk_management | usability | ifu | pmap | methods | security_risk | other
version	Document version
summary_text	User-provided summary of the document's key content (for AI drafting)
sort_order	Display order in the references table

Auto-populated references: Documents from CORTEX modules are auto-added:

- [R1] SOA document → from linked SOA
- [R2] Standalone Validation Report → from linked Validation
- [R3] Clinical Benefit Report → from linked Validation (if MRMC exists)
- [R11] PMCF Plan → from linked PMS
- [R12] PMS Plan → from linked PMS
- [R13] Named Device Literature Search Report → from linked SLS
- [R14] PCCP → from linked PCCP (if exists)

Manual references: The user adds external documents (Risk Report, Usability, IFU, PMAP, Methods Document) with their identifiers and summary text.

The references table is auto-compiled and included in the CER export front matter.

4.7 Evaluator Records

Field	Description
id	UUID
cer_analysis_id	Parent CER
role	written_by | verified_by | approved_by
full_name	e.g., "Vladimir Laletin MD, PhD"

Field	Description
title	e.g., "Clinical and Scientific Research Engineer"
affiliation	e.g., "Avicenna.AI"
date	Signature date
cv_attached	Boolean — whether CV is uploaded
conflict_of_interest	Text — declaration of COI
sort_order	Display order

The evaluator table is auto-populated in CER §9 and in the document header.

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5. GSPR Compliance Section

5.1 Concept

The GSPR section (CER §2) is a structured table mapping each relevant General Safety and Performance Requirement to the clinical evidence supporting conformity. It is auto-populated from two sources:

1. **Validation module** → GSPR mapping matrix (endpoints → GSPRs → evidence → status)
2. **External Risk Management** → risk-related GSPRs (user confirms coverage)

5.2 GSPR Record (CER-level)

Field	Description
gspr_number	e.g., "1", "4", "6", "8", "17.1"
gspr_text	Full requirement text (from MDR Annex I)
evidence_summary	How this GSPR is addressed — auto-generated from Validation mapping
source_type	validation | risk_management | usability | soa | manual
source_reference	Which document/module provides the evidence
status	confirmed | partially_addressed | not_addressed
notes	Free-form for additional context

5.3 Default GSPR List for SaMD Class IIa/IIb

Pre-loaded and configurable. Default includes: §1, §4, §6, §8, §17.1, §17.2 (as applicable). Users can add/remove GSPRs per project.

5.4 CER §7 GSPR Compliance Statement

The Discussion & Conclusions section (§7) includes an auto-generated GSPR compliance statement:

> "The clinical evaluation of CINA-CSpine complies with Regulation (EU) 2017/745, and the conformity with General Safety and Performance Requirements (GSPRs 1, 4, 6, 8, 17.1) was demonstrated."

This statement is auto-compiled from the GSPR table — listing only GSPRs with status confirmed.

---

6. Benefit-Risk Determination

6.1 Concept

CER §6 is a structured benefit-risk analysis combining:

- **Benefit Registry** from the Validation module (BL calculations)
- **Risk Summary** from the external Risk Management file (RL values — user-entered)

This section uses the **MDR BL/RL scored matrix framework**. FDA submissions use a qualitative benefit-risk assessment per FDA's Benefit-Risk Guidance for Premarket Submissions — this is handled in the 18.CVS (Validation module), not in the CER. CORTEX does not attempt to map between the two frameworks — each regulatory context uses its native benefit-risk methodology.

6.2 Risk Summary (User-Entered)

Since Risk Management is external to CORTEX v1, the user provides:

Field	Description
total_risks_identified	e.g., 24
risks_by_category	e.g., "10 design, 5 production, 9 usability"
pre_mitigation_summary	e.g., "11 low, 8 medium, 5 high"
post_mitigation_summary	e.g., "23 low, 1 medium, 0 high"
max_residual_rl	e.g., 6
max_residual_rl_offlabel	e.g., 9 (off-label use scenario)
key_residual_risks	Structured list of residual risks with descriptions
overall_residual_risk_statement	e.g., "All risks fell to acceptable level..."
risk_document_ref	Reference to external Risk Management Report

6.3 Benefit-Risk Computation

For each benefit in the Validation module's benefit registry:

> Benefit Level (BL) = Probability Score (1-5) × Magnitude Score (1-5)

The CER auto-computes:

- **Overall BL** = average or max of all individual BLs (configurable)
- **Benefit/Risk Ratio** = BL / RL
- **Acceptability:** ratio ≥ 1 → acceptable; ratio < 1 → flagged

For off-label scenarios:

- **Off-label Ratio** = BL / RL_offlabel

6.4 Auto-Generated Content

The system generates:

1. **Benefit-Risk Summary Table** (benefits left, risks right — matching the CER format)
2. **Magnitude Scale Table** (5-point scale with definitions — configurable)
3. **Per-benefit analysis sub-section** (one per benefit, with BL calculation table)
4. **PCCP comparison** (when linked): planned benefit targets vs. achieved results, with deviation analysis
5. **Benefit/Risk Conclusion paragraph:** "We conclude that the benefits (BL=X) of [device] outweigh the minimal potential residual risks (RL=Y)..."

All auto-generated content is editable by the user.

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7. Named Device Literature Search

7.1 Concept

CER §5.3 requires a systematic literature search for published data about the named device itself (not similar devices — that's the SOA). This is managed as a dedicated SLS session.

7.2 Implementation

- SLS session type: `named_device`
- Scope: literature mentioning the specific device (e.g., "CINA-CSpine" + "Avicenna.AI")
- Databases: scientific databases (PubMed, Scholar, Cochrane, PMC, ClinicalTrials) + competent authority databases (MAUDE, ANSM, BfArM, AFMPS, etc.)
- Time period: from first installation in near-clinical settings to date
- Keywords: device name, manufacturer name, product variants
- Full SLS workflow: query → execute → screen → include/exclude → PRISMA

7.3 Integration with CER

- The Named Device SLS is linked to the CER via `linked_sls_named_device_id`
- CER §5.3 is auto-drafted from the SLS results:
  - If articles found: summary of each included article
  - If no articles found: "No relevant clinical data related to [device] was found in the current systematic literature search."
- The SLS search strategy and PRISMA flowchart are included as a CER appendix or within §5.3

7.4 Annual Update

For CER updates (annual or patch), the named device SLS session is duplicated with updated date range. The new session goes through the full SLS workflow. Delta analysis shows new articles found since the previous search.

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8. CER Versioning

8.1 Version Types

Type	Trigger	Scope
initial	First CE marking	Full CER, all sections
annual_update	Annual PMS cycle (PSUR) triggers re-evaluation	Focus on: new PMS data, literature update, benefit-risk reassessment
patch_update	Software patch release	Focus on: updated Validation results, delta analysis, minimal narrative changes

8.2 Creating a New Version

When the user creates a new CER version:

1. The system duplicates the previous locked CER
2. The new CER is in `drafting` status
3. The user can:
    - Re-link to updated upstream modules (new SOA version, new Validation study, new SLS)
    - Update external document references (new Risk Report version, etc.)
    - Link to a PMS cycle (for annual updates)
4. Sections that haven't changed retain their content (with a "unchanged since v[N]" indicator)
5. Sections that need updating are flagged based on what upstream data changed

8.3 Delta Tracking

For `annual_update` and `patch_update` versions, the system tracks:

- Which upstream modules changed (new SOA, new Validation, new SLS results)
- Which CER sections are affected by the changes
- A delta summary auto-generated: "CER v02 updates: §5.2.3.2 (new PMS data), §6 (benefit-risk re-assessed), §5.3 (literature update with 2 new articles found)"

8.4 Lock Behavior

- Locking a CER also locks (or verifies lock of) all linked upstream modules
- A locked CER is read-only — creating a new version is the only way to make changes
- Unlocking requires manager role + documented justification

9. Workflow & Statuses

9.1 Lifecycle

planned → drafting → review → completed → locked

Status	What Happens	What's Editable	Export
planned	CEP + PCCP being written: structure defined, methodology planned, targets set	Everything	CEP (DOCX), PCCP (DOCX)
drafting	CER being written: AI drafting sections, linking upstream data	Everything	Draft CER (DOCX, watermarked)
review	All sections drafted, under human review	Content only (no structural changes)	Review CER (DOCX)
completed	CER finalized, ready for submission	Minor edits (logged)	Final CER (DOCX)
locked	Frozen for regulatory submission. Read-only.	Nothing	Final CER (DOCX)

9.2 Section-Level Workflow

Each section has its own status (empty → planned → drafted → reviewed → final), allowing parallel work across sections. The CER-level status is the minimum of all section statuses (with warnings for lagging sections).

9.3 Transition Validations

planned → drafting:

- All upstream modules must be linked
- At least SOA Clinical and one Validation study must be completed

drafting → review:

- All sections must be at least drafted
- GSPR table must be populated
- Benefit-risk section must be computed
- Bibliography must be compiled

review → completed:

- All sections must be reviewed or final
- No unresolved warnings (e.g., missing external document summaries)

completed → locked:

- All evaluator records must be filled (written_by, verified_by, approved_by)
- All upstream modules must be locked

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10. Exports

Export	Format	Content
Clinical Evaluation Plan (CEP)	DOCX	CER in planned status — prospective language, TBD placeholders, methodology description
Performance and Clinical Claims Plan (PCCP)	DOCX	Planned claims with SOA justification, validation endpoint mapping, traceability matrix
Clinical Evaluation Report (20.CER)	DOCX	Full report with all sections, tables, cross-references, bibliography
GSPR Compliance Matrix	XLSX / DOCX	GSPRs × evidence mapping (standalone export)
Benefit-Risk Summary	XLSX / DOCX	Benefit registry + risk summary + BL/RL computation
PCCP Comparison Report	XLSX / DOCX	Planned claims vs. achieved results with deviation analysis
References Table	XLSX	All project references with identifiers and titles
Bibliography	DOCX / RIS	All cited articles compiled with full references
CER Update Delta	DOCX	Changes between CER versions (for annual/patch updates)
Full CER Package	ZIP	CER + all appendices

10.1 Full CER Package — Appendix Structure

The CER DOCX export includes the main body. Appendices are separate documents (exported from their respective modules) but linked via the references table:

```
CER Package:
├── CER Main Body (DOCX) — document ID: 20.CER
├── Appendix A: State of the Art (SOA) — from SOA module export
├── Appendix B1: Standalone Validation Report — from Validation module export
├── Appendix B2: Clinical Benefit Report — from Validation module export (if MRMC)
├── Appendix C: Evaluator Qualifications (CVs) — uploaded documents
├── Appendix D1: Literature Search Report (SOA) — from SLS module export
├── Appendix D2: Named Device Literature Search Report — from SLS module export
├── Appendix E: PCCP with Comparison Report (if exists)
└── (External documents referenced but not included:
    Risk Mgmt, Usability, IFU, etc.)
```

Note: The 18.CVS (FDA Clinical Validation Study) is NOT included in the CER package. It is a parallel FDA deliverable produced by the Validation module.

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11. Async Operations

Operation	Estimated Duration	Async?
AI section drafting (single)	10-30s	No (inline with spinner)
AI section drafting (all sections)	10-30s × N sections	Yes
Bibliography compilation	2-5s	No (inline)
GSPR matrix auto-population	2-5s	No (inline)
Benefit-risk computation	< 1s	No (inline)
CER export (DOCX)	10-30s	No (inline with spinner)
CEP export (DOCX)	5-15s	No (inline with spinner)
PCCP export (DOCX)	5-15s	No (inline with spinner)
Full package export (ZIP)	30-90s	Yes

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12. Permissions

Role	Can Do
clinical / regulatory	Full access: create CER, draft sections, link modules, review, complete

Role	Can Do
manager	Full access + lock/unlock CER with justification, approve evaluator records
collaborateur	Read-only access to completed/locked CER and exports
admin	Everything + manage section templates at system level

13. Open Items

1. **Regulatory template variants:** The default template matches Avicenna.AI's current format (MEDDEV 2.7/1 Rev 4). For 510(k) submissions, a different CER structure may be needed (closer to a Clinical Summary). The template system should support multiple base templates per regulatory context. Note: FDA clinical validation evidence is now handled via the 18.CVS in the Validation module, reducing the need for a full FDA CER variant.
2. **CEP-to-CER transition automation:** When transitioning from planned to drafting, the system could auto-detect which CEP sections already have upstream data available and offer to auto-draft them. This needs UX refinement. The PCCP should be automatically archived at this transition point.
3. **Cross-reference numbering:** The CER uses [R1], [R2], etc. for document references and [1], [2], etc. for bibliography. Auto-numbering must handle both systems without conflicts. Reference numbers should auto-update when references are added or reordered.
4. **External document version tracking:** When external documents (Risk Report, Usability) are updated, the CER should detect the version mismatch and flag affected sections for review.
5. **Multi-product CER:** Some products share clinical context (same pathology, same SOA Clinical). The system should support linking multiple products to a shared SOA Clinical while maintaining separate SOA Device and Validation tracks. This is a future enhancement.
6. **DOCX formatting fidelity:** The CER export must match the exact formatting of existing Avicenna.AI templates (headers, table styles, cover page, signature block). A DOCX template engine with configurable styles is needed.
7. **Bibliography management:** The CER bibliography is compiled from all articles referenced across all sections (SOA + Validation + SLS). Duplicate detection and consistent formatting (Vancouver style) across modules is essential. The system should support both in-text numbered references [1] and author-year references [Author, Year] — configurable per project.
8. **PCCP deviation workflow:** When post-validation results deviate from PCCP planned claims, the system flags deviations but does not enforce corrective actions. The clinical affairs team decides whether the deviation requires PCCP amendment, CER discussion justification, or acceptance. The threshold for "significant deviation" is configurable per project (e.g., >5% absolute difference from planned target triggers a flag).

9. **PCCP versioning across CER updates:** When a CER is updated (annual or patch), the PCCP may need updating if new claims are added or existing claims are revised based on new evidence. The system should maintain PCCP version history linked to CER versions, showing how planned claims evolved over time.

Version History

Version	Date	Changes
v1.0	January 2026	Initial specification
v1.1	February 2026	Regulatory gap analysis, GSPR compliance section, benefit-risk framework
v1.2	February 2026	Internal review integration: 20.CER/18.CVS separation, PCCP addition, IMDRF SaMD classification reference, BL/RL MDR-only framework clarification, PCCP traceability matrix, CER package structure update